

TECHNICAL SPECIFICATION SHEET

NS1NANOSYNTH MNMOD V1.2.0

Firmware Version: v1.2.0

Target Hardware: soundmachines NS1nanosynth (ATmega32U4 Core)

Classification: Hardware Reference Manual & Pin Assignment Guide

1. System Overview & Core Outputs

This section details the primary digital communication buses, hardware control triggers, and foundational gate outputs managed by the v1.2.0 firmware.

Pin	Electrical Type	Signal Name	Functional Description
Pin 4	Digital Output	SPI_SS	MCP4922 DAC Slave Select. Driven LOW to initiate SPI DAC transmission.
Pin 5	Digital Output	GATE_OUT	Note Gate Signal. Generates +5V when a MIDI Note-On is active; drops to 0V on Note-Off.
Pin 12	Digital Output	TRIG_OUT	Master Clock Sync. +5V trigger pulse, fixed 15ms duration, fired every 1/16th Note .
Pin 2 / 3	I2C Data/Clock	SDA / SCL	Serial bus for the external 4-channel Digital Potentiometer (Address 0x2C).

2. Dedicated LFO Engine (Pin 11)

The LFO architecture outputs natively on **Pin 11 (PWM)**. Waveform geometry and clock subdivisions are determined entirely by grounding dedicated input matrices.

2.1 Waveform Selection Matrix (Pins 0 & 1)

Internal pull-up resistors are active. Ground the designated pins to select the active waveform. Engine processes math logic in real-time unless stated otherwise.

Pin 1 State	Pin 0 State	Binary Logic	Active Waveform (Pin 11)
GND	GND	00	Inverse Sawtooth

GND	Open	01	Sawtooth
Open	GND	10	Square / Pulse (50% Duty)
Open	Open	11	Sine Wave (256-Step PROGMEM Table)

2.2 LFO Subdivision Matrix (Pins 6, 7 & 8)

Subdivisions are calculated against incoming Master MIDI Clock messages (24 PPQN). The firmware logic actively inverts the pull-up state (GND = Active/1).

Pin 6 (Mod 1)	Pin 7 (Mod 2)	Pin 8 (Mod 3)	Musical Division	MIDI Clock Ticks
Open	Open	GND	1/8 Note	12 Ticks
Open	GND	GND	1/4 Note Triplet	16 Ticks
GND	Open	GND	Dotted 1/8 Note	18 Ticks
Open	Open	Open	1/4 Note (Quarter)	24 Ticks
GND	GND	GND	1/2 Note Triplet	32 Ticks
GND	GND	Open	Dotted 1/4 Note	36 Ticks
Open	GND	Open	1/2 Note (Half)	48 Ticks
GND	Open	Open	1/1 Note (Whole)	96 Ticks

3. Performance & Control Voltage Matrices

3.1 Portamento (Glide) Matrix

Grounding combinations determines the glide time coefficient between consecutive overlapping MIDI notes.

Pin A2	Pin A1	Pin A0	Pitch & CV Glide Time
Open	Open	Open	Disabled (Instantaneous)
Open	Open	GND	50 ms
Open	GND	Open	100 ms
GND	Open	Open	250 ms
Any 2-Pin configurations: Open / GND / GND GND / Open / GND GND / GND / Open			500 ms
GND	GND	GND	1000 ms (1 Second)

3.2 Analog CV to MIDI CC Conversion

Analog inputs sample control voltages and stream them over USB MIDI. Handled via Exponential Moving Average (EMA) and deadband hysteresis to eliminate line noise.

Analog Pin	Target Destination	DSP Processing Stack
A3	MIDI CC102	10-bit Oversampling + Hysteresis Filter
A4	MIDI CC103	10-bit Oversampling + Hysteresis Filter
A5	MIDI CC104	10-bit Oversampling + Hysteresis Filter

Hardware Note: In version 1.2.0, pins A1 and A2 are isolated from the CV scanning block and reserved exclusively for the Portamento matrix. Digital input buffers for A3, A4, and A5 are hardware-disabled via `DIDR0` to ensure a pristine analog read.

4. Peripheral Digital Potentiometer Control

Incoming MIDI CC messages actively scale the resistance profiles of the external 4-channel digital potentiometer via I2C.

Incoming MIDI CC	Target Output Node	I2C Address Space
CC30	Potentiometer Channel 1	0x00
CC31	Potentiometer Channel 2	0x10
CC32	Potentiometer Channel 3	0x60
CC33	Potentiometer Channel 4	0x70